Machine Learning for Many-Body Physics

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https://wangleiphy.github.io

Recommender Engines for QMC

Li Huang and LW, 1610.02746
Li Huang, Yi-feng Yang and LW, 1612.01871

A Comparison of two Markov Chain Monte Carlo samplers

What is machine learning?

"Give computers the ability to learn without being explicitly programmed."

—Arthur Samuel





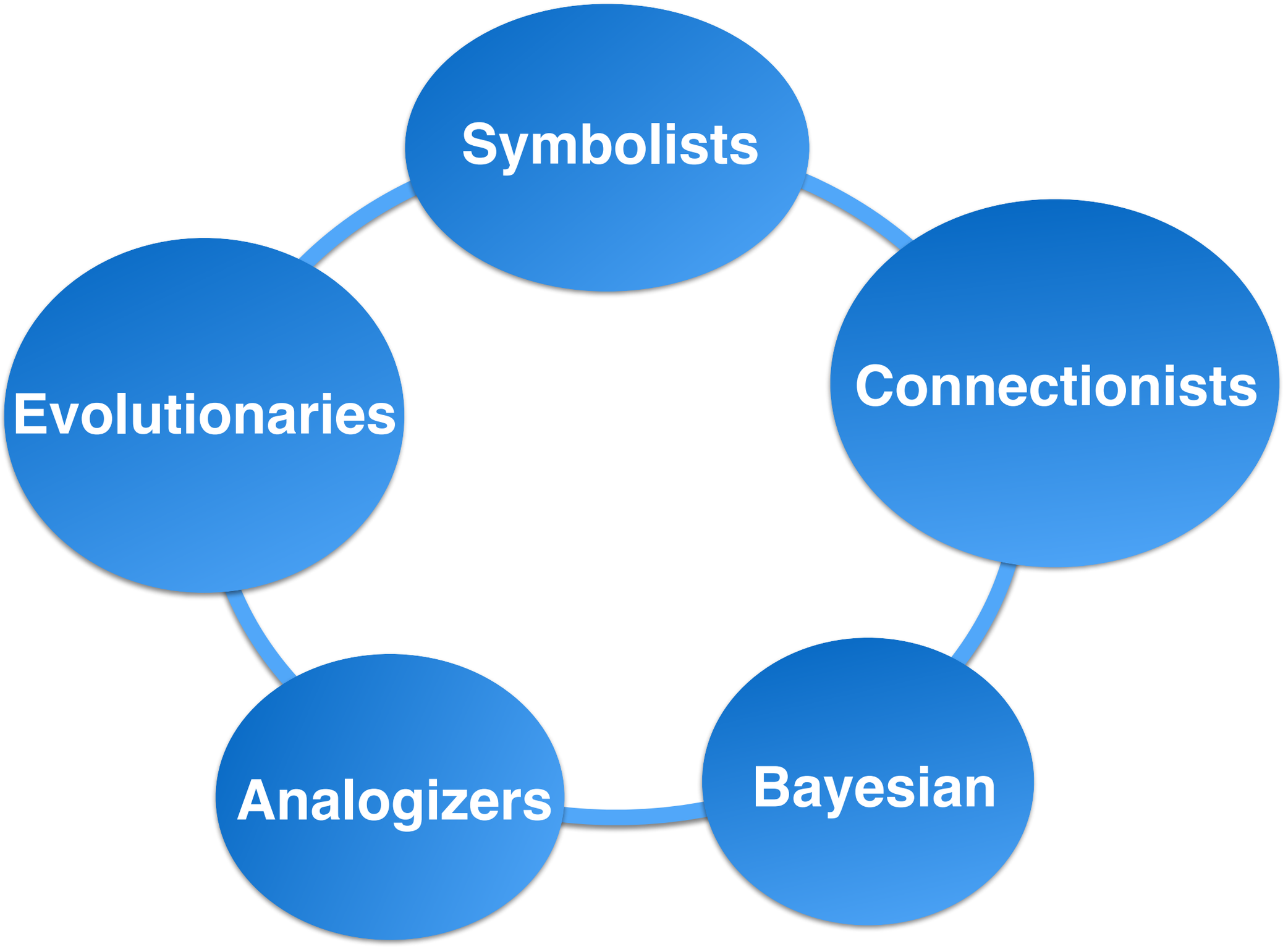
Machine Learning



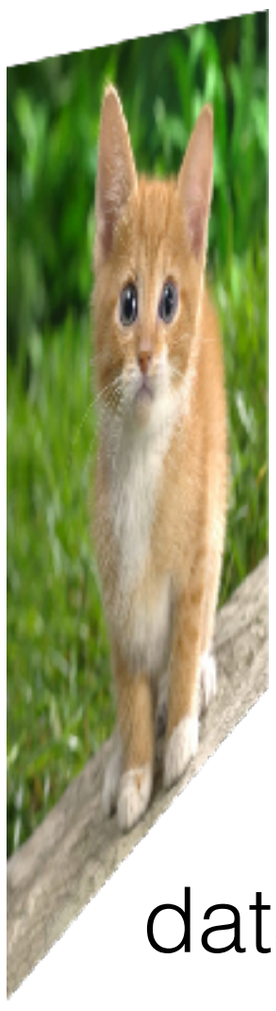


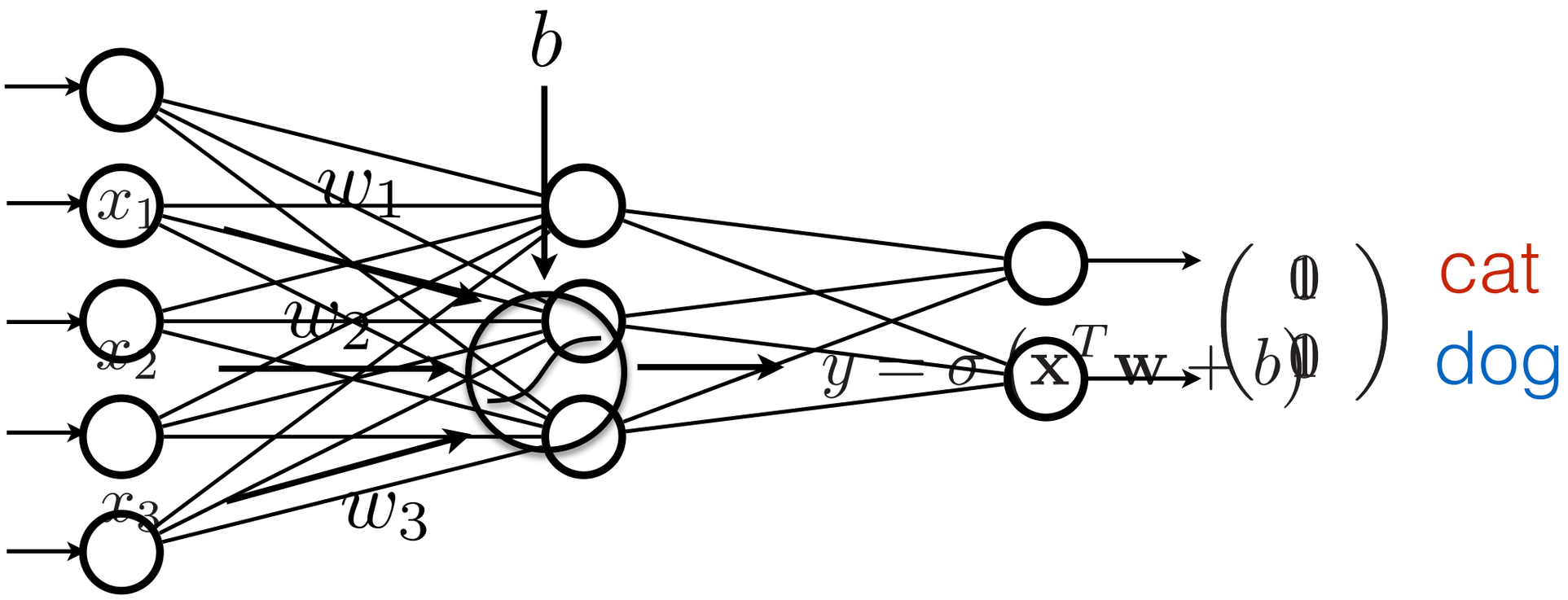


Five schools of ML



Neural networks





Computing unit: artificial neuron

data

label

Why it works? Not only a math problem.

Universal Function Approximator

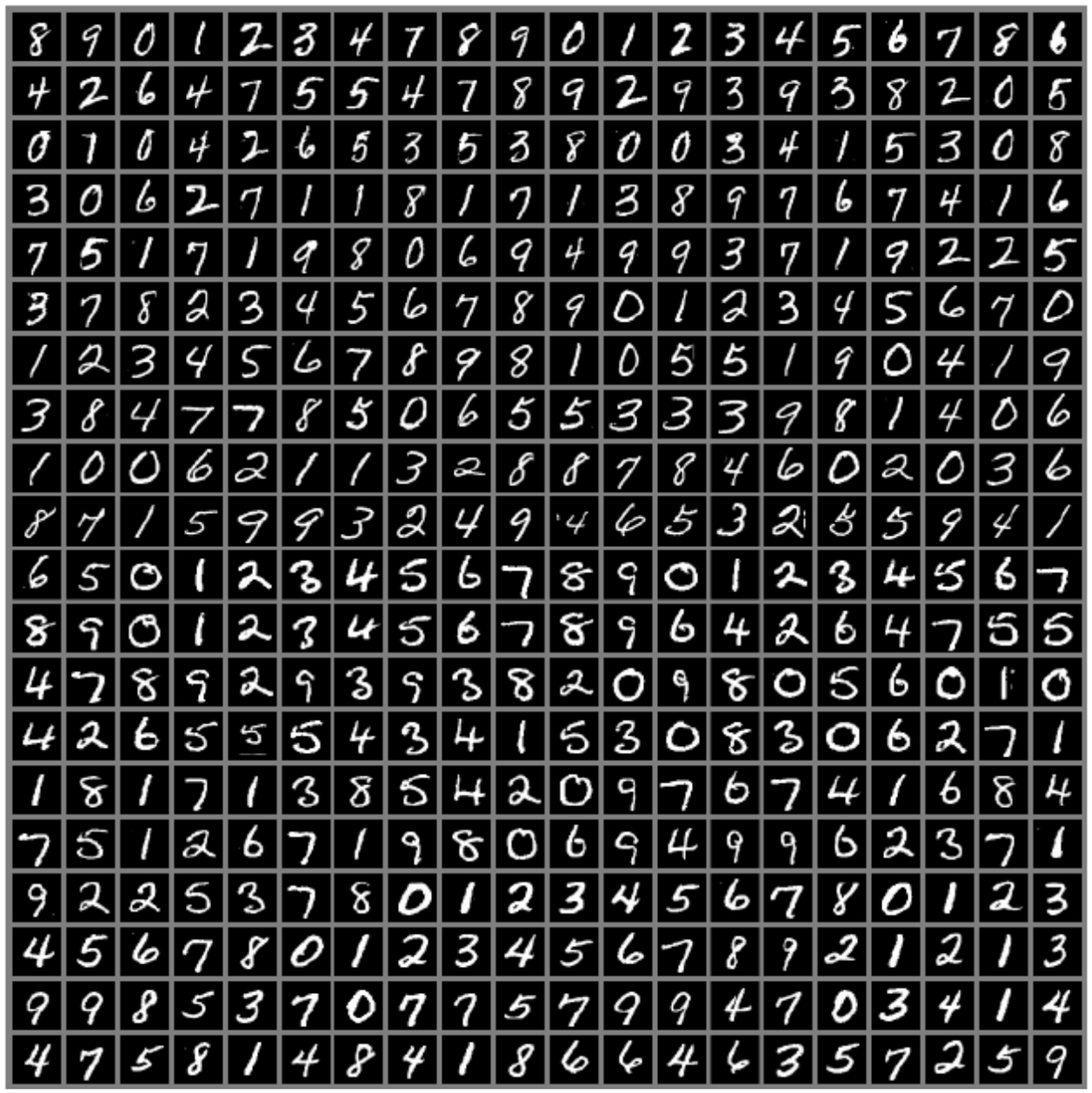
It is because of the law of physics: symmetry, locality, compositionality and low-order polynomials

Cybenko 1989

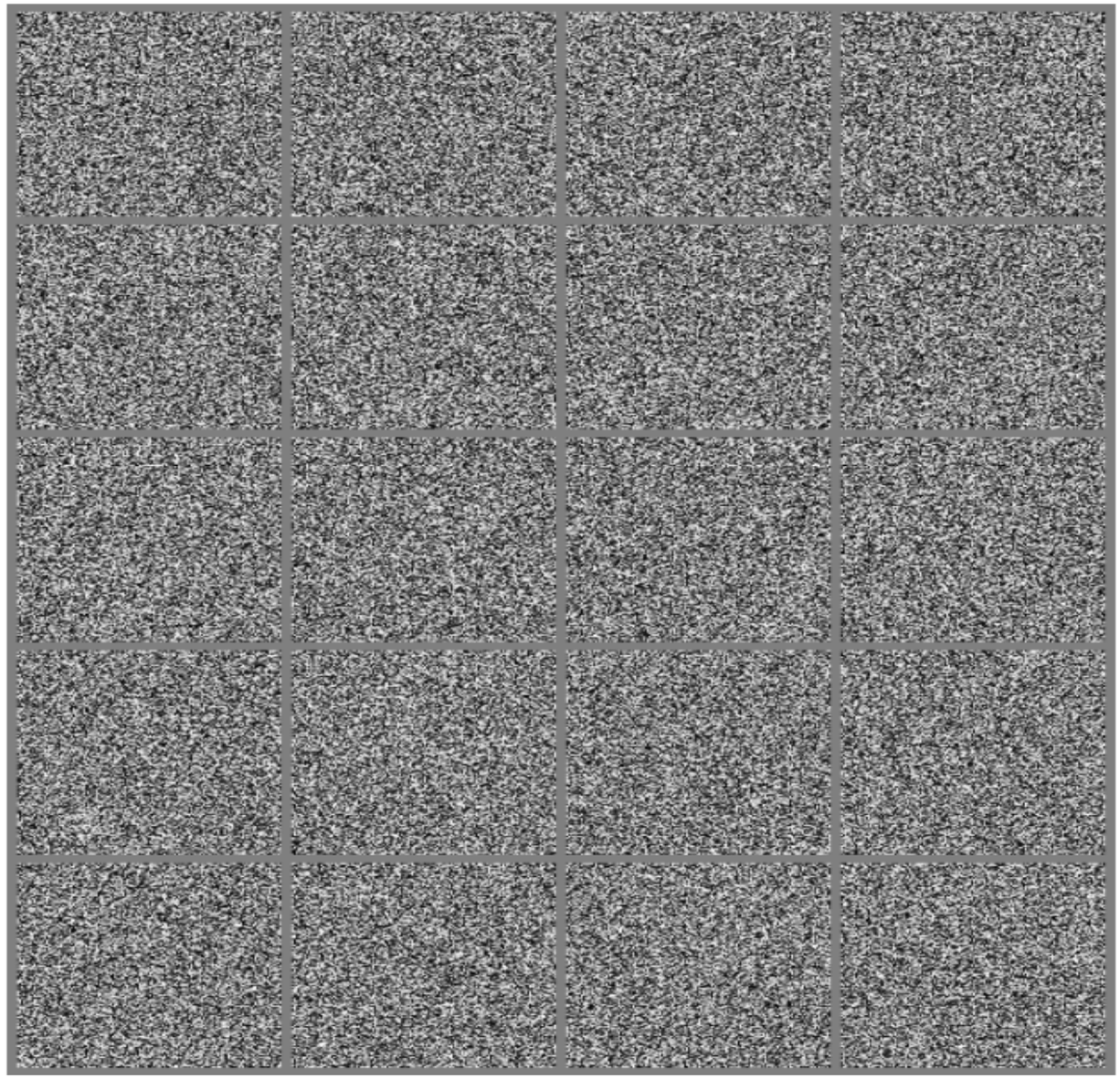
Horniket al. 1989

Lin, Tegmark, 1608.08225

MNIST database



random images

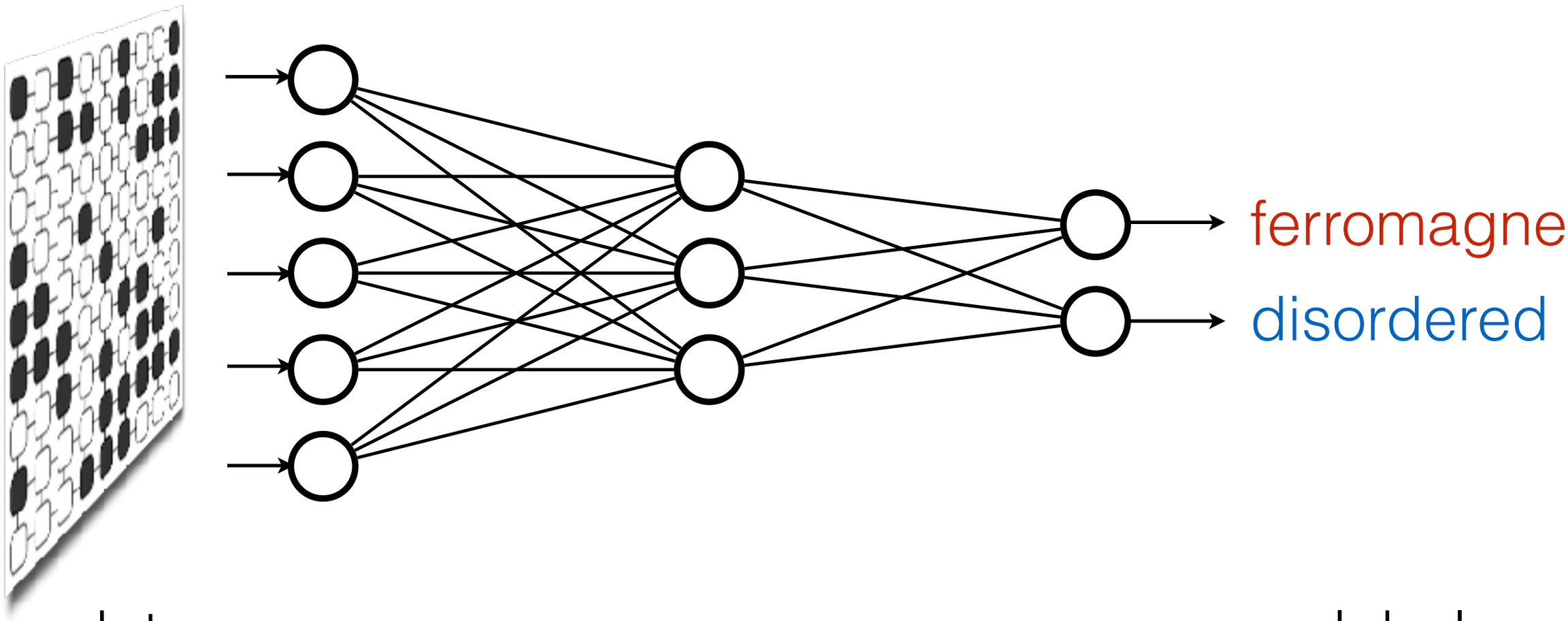


from the “Deep Learning” book by Goodfellow, Bengio, Courville

https://www.deeplearningbook.org/

Machine learns phases of matter

Ising configurations



Carrasquilla and Melko, 1605.01735

Machine learns everything…

Broecker, Carrasquilla, Melko, Trebst, 1608.07848

Ch'ng, Carrasquilla, Melko, Khatami, 1609.02552

Tubman,1609.08142

Tanaka, Tomiya 1609.09087

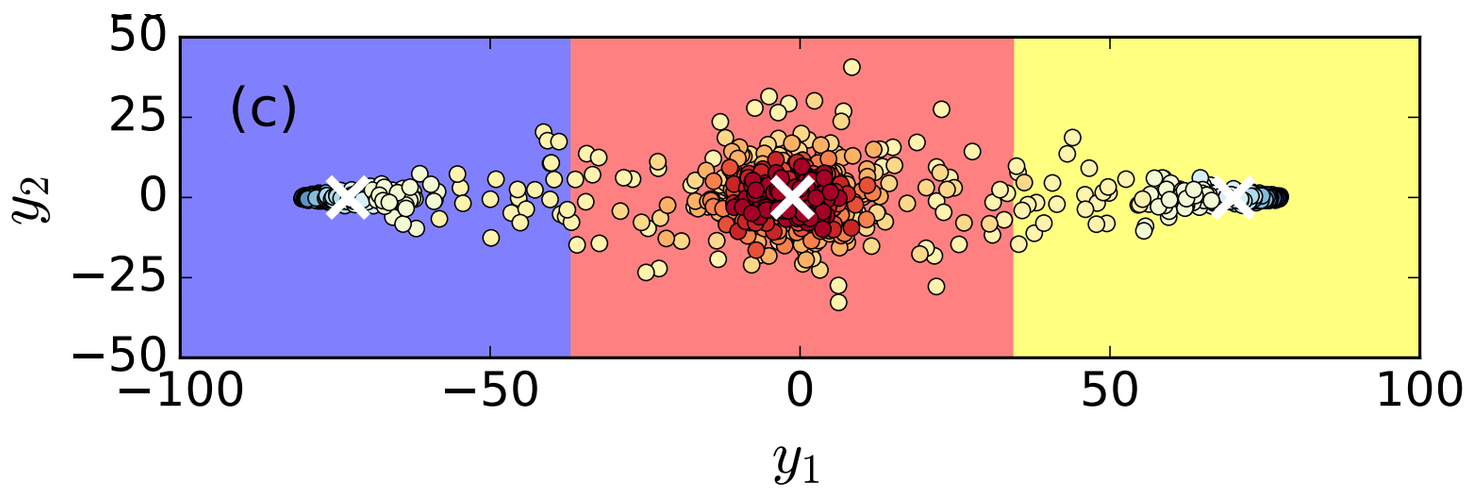
Ohtsuki, Ohtsuki,1610.00462

Zhang, Kim,1611.01518

- However, needs labels for supervised learning

- Also, often needs hand-crafted features

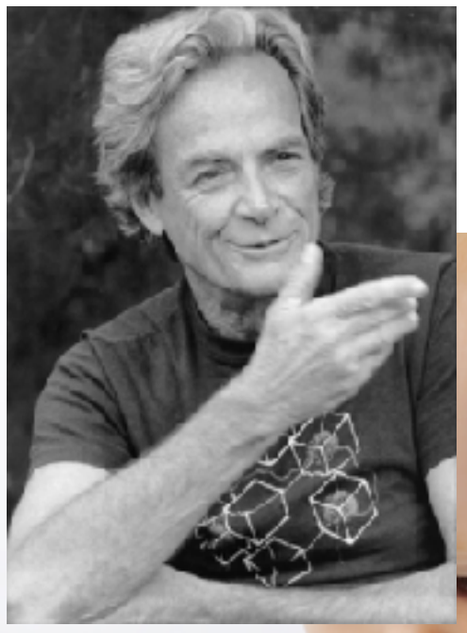
- A way out: unsupervised feature learning

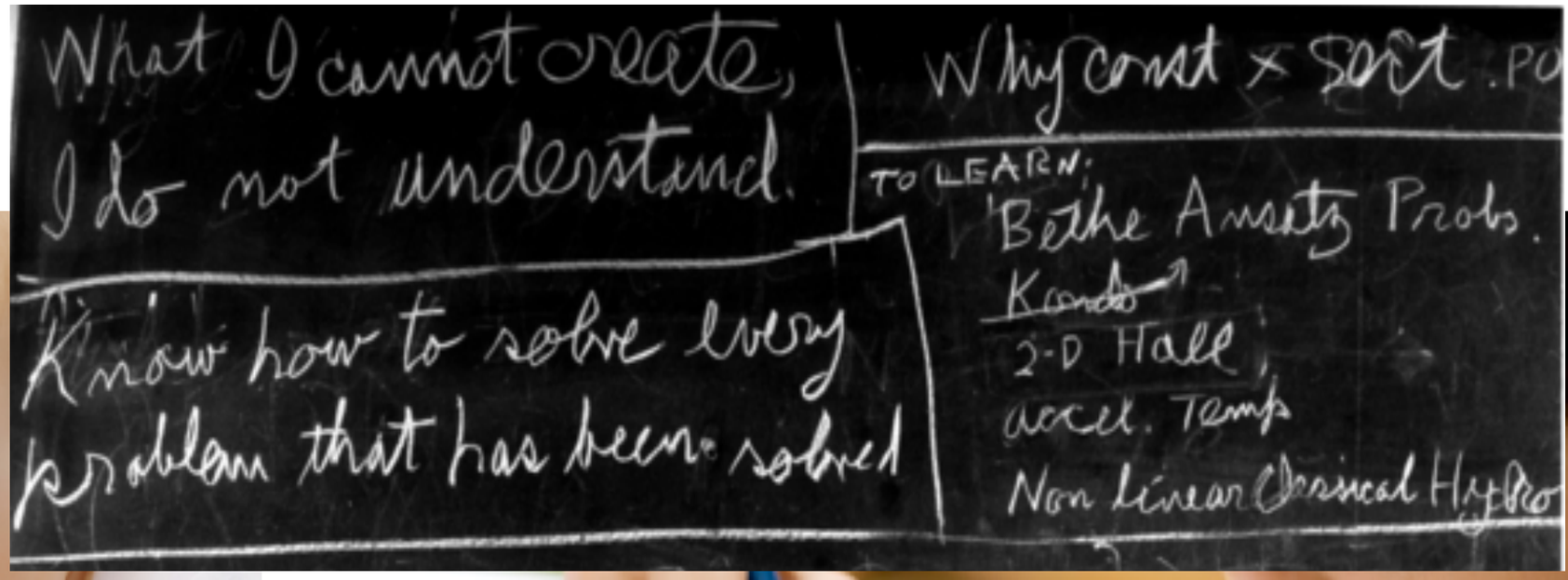


"Discovering Phase Transitions with Unsupervised Learning"

LW, 1606.00318

Discriminative vs Generative Learning







"What I can not create, I do not understand"

read

ELSEVIER

Progress in Brain Research

Volume 165, 2007, Pages 535–547

Computational Neuroscience: Theoretical Insights into Brain Function



To recognize shapes, first learn to generate images

Geoffrey E. Hinton

Department of Computer Science, University of Toronto, 10 Kings College Road, Toronto, M5S 3G4 Canada



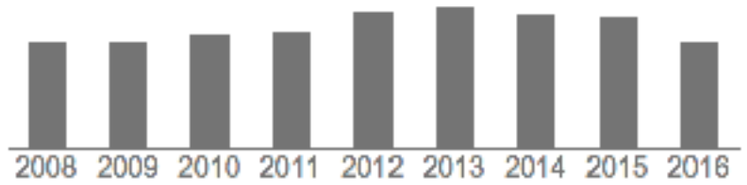
Richard Feynman

California Institute of Technology
quantum mechanics, quantum electrodynamics
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The Feynman lectures on physics RP Feynman, RB Leighton, M Sands, SB Treiman Physics Today 17, 45	13294 *	1964
Simulating physics with computers RP Feynman International journal of theoretical physics 21 (6), 467-488	5367	1982
Space-time approach to non-relativistic quantum mechanics RP Feynman Reviews of Modern Physics 20 (2), 367	4224	1948
Forces in molecules RP Feynman Physical Review 56 (4), 340	3289	1939
There's plenty of room at the bottom RP Feynman Engineering and Science 23 (5), 22-36	3109	1960
Very high-energy collisions of hadrons RP Feynman Physical Review Letters 23 (24), 1415-1417	2748	1969
Theory of the Fermi interaction RP Feynman, M Gell-Mann Physical Review 109 (1), 193	2632	1958
The character of physical law R Feynman MIT press	2449	1967



Geoffrey Hinton

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machine learning, neural networks, artificial intelligence, cognitive science, computer science
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Title 1–20	Cited by	Year
Parallel distributed processing DE Rumelhart, JL McClelland, PDP Research Group IEEE 1, 354-362	20861	1988
Learning internal representations by error-propagation DE Rumelhart, GE Hinton, RJ Williams Parallel Distributed Processing: Explorations in the Microstructure of ...	19525	1986
Learning internal representations by error propagation DE Rumelhart, GE Hinton, RJ Williams CALIFORNIA UNIV SAN DIEGO LA JOLLA INST FOR	~~19519~~	1985
Learning representations by back-propagating errors DE Rumelhart, GE Hinton, RJ Williams Nature 323, 533-536	10966	1986
Imagenet classification with deep convolutional neural networks A Krizhevsky, I Sutskever, GE Hinton Advances in neural information processing systems, 1097-1105	7876	2012
Parallel distributed processing JL McClelland, DE Rumelhart, PDP Research Group MIT press	5150	1987
A fast learning algorithm for deep belief nets GE Hinton, S Osindero, YW Teh Neural computation 18 (7), 1527-1554	4873	2006
Reducing the dimensionality of data with neural networks GE Hinton, RR Salakhutdinov Science 313 (5786), 504-507	4047	2006

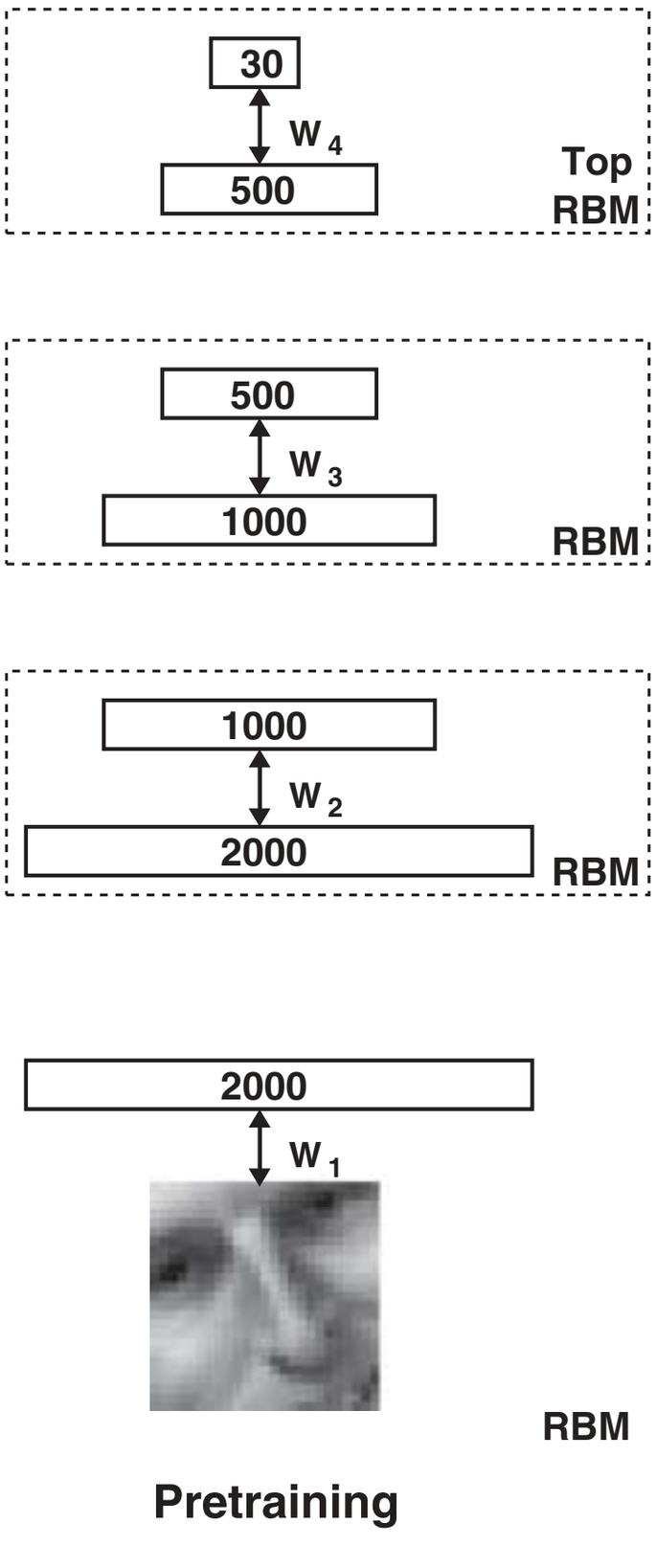
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Co-authors View all...

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Chris Williams
Vinod Nair
Sidney Fels
David C. Plaut
Robert Tibshirani
Demetri Terzopoulos
Steven L. Small
Rama Chellappa



Reducing the Dimensionality of Data with Neural Networks

G. E. Hinton* and R. R. Salakhutdinov

High-dimensional data can be converted to low-dimensional codes by training a multilayer neural network with a small central layer to reconstruct high-dimensional input vectors. Gradient descent can be used for fine-tuning the weights in such "autoencoder" networks, but this works well only if the initial weights are close to a good solution. We describe an effective way of initializing the weights that allows deep autoencoder networks to learn low-dimensional codes that work much better than principal components analysis as a tool to reduce the dimensionality of data.

Dimensionality reduction facilitates the classification, visualization, communication, and storage of high-dimensional data. A simple and widely used method is principal components analysis (PCA), which finds the directions of greatest variance in the data set and represents each data point by its coordinates along each of these directions. We describe a nonlinear generalization of PCA that uses an adaptive, multilayer "encoder" network

2006 VOL 313 **SCIENCE** www.sciencemag.org

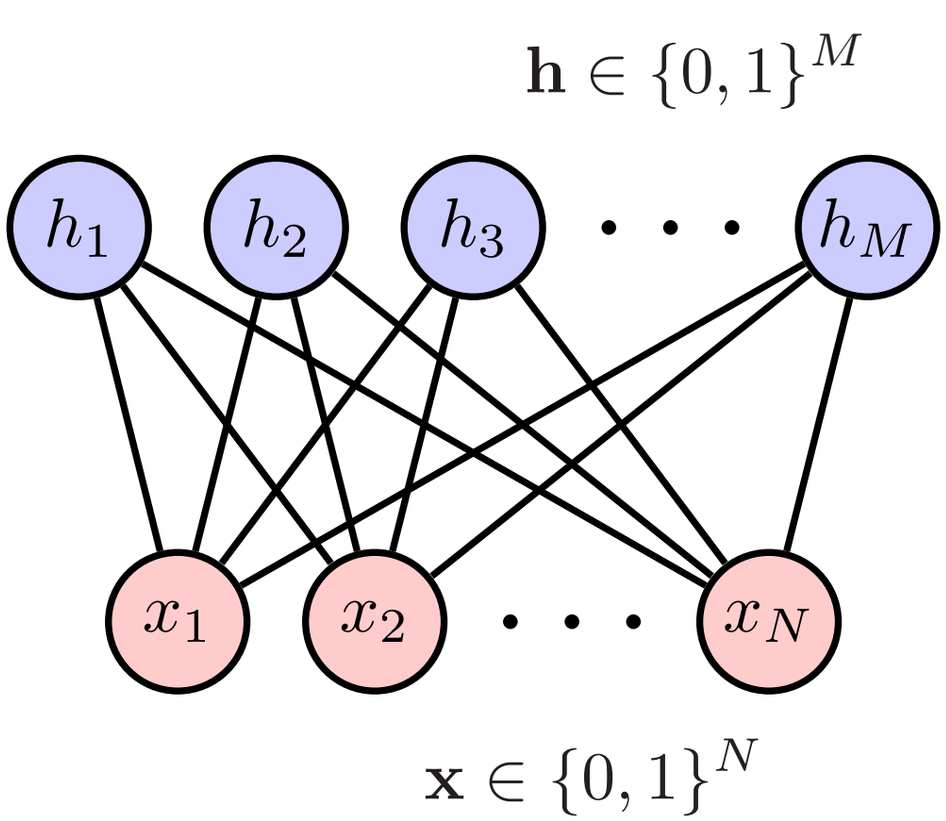
Renaissance of deep learning

Restricted Boltzmann Machine

Smolensky 1986

$$E(\mathbf{x}, \mathbf{h}) = -\sum_{i=1}^{N} a_i x_i - \sum_{j=1}^{M} b_j h_j - \sum_{i=1}^{N}\sum_{j=1}^{M} x_i W_{ij} h_j$$

Energy based model



Probabilities

$$p(\mathbf{x}, \mathbf{h}) \sim e^{-E(\mathbf{x},\mathbf{h})}$$

$$p(\mathbf{x}) = \sum_{\mathbf{h}} p(\mathbf{x}, \mathbf{h})$$

$$p(\mathbf{h}|\mathbf{x}) = p(\mathbf{x}, \mathbf{h})/p(\mathbf{x})$$

Universal approximator of probability distributions

. Freund and Haussler, 1989 Le Roux and Bengio, 2008

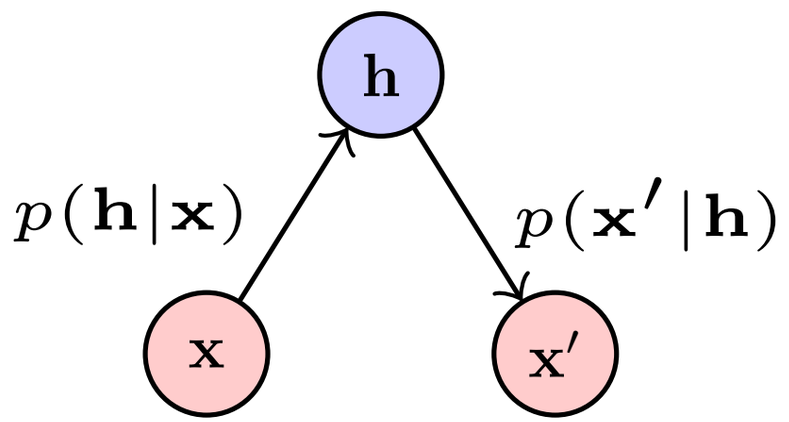
Training & Sampling

Contrastive divergence Hinton 2002

$$\Delta W_{ij} = \eta \left(\langle x_i h_j \rangle_{\text{data}} - \langle x_i h_j \rangle_{\text{model}} \right)$$

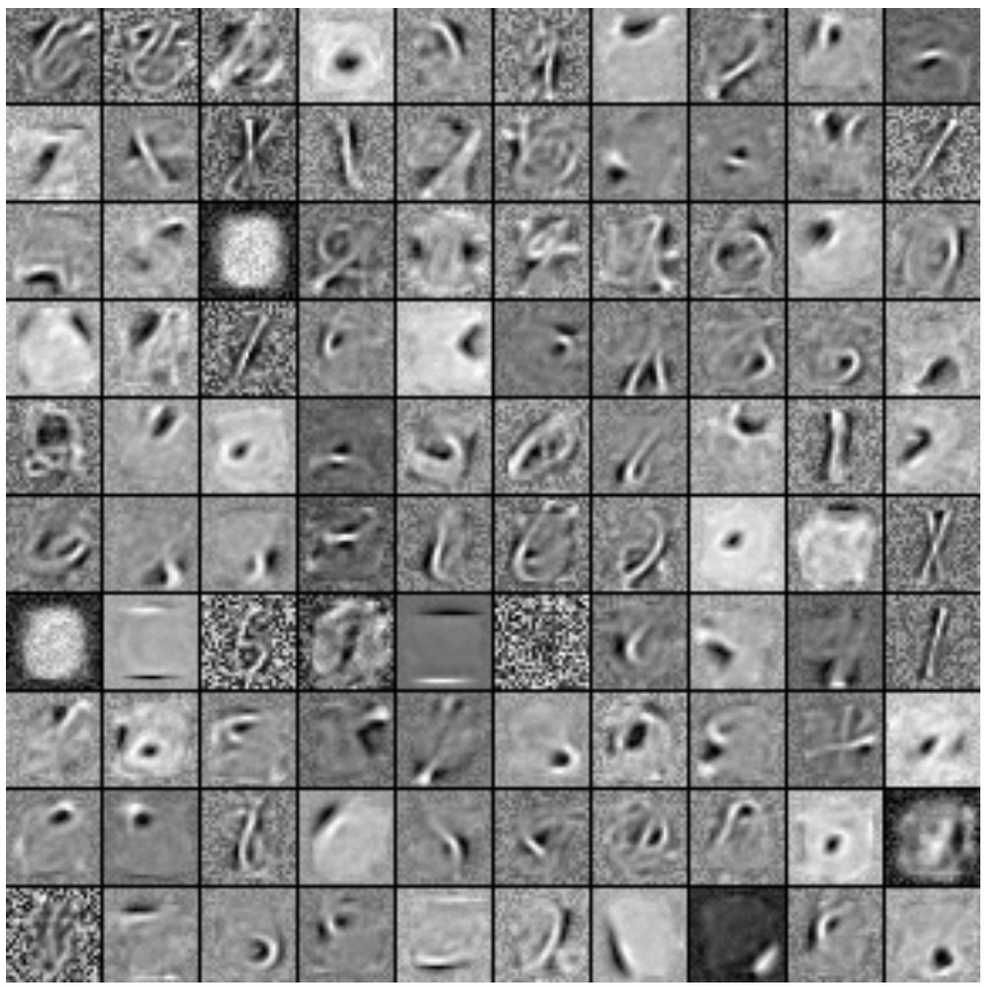
determining the model given the data

Blocked Gibbs sampling



generate more data using the learned model





MNIST data

learned weight W_{ij}

784 100

Theano deep learning tutorial http://www.deeplearning.net/tutorial/rbm.html#rbm

Challenge

Which one is written by human ?



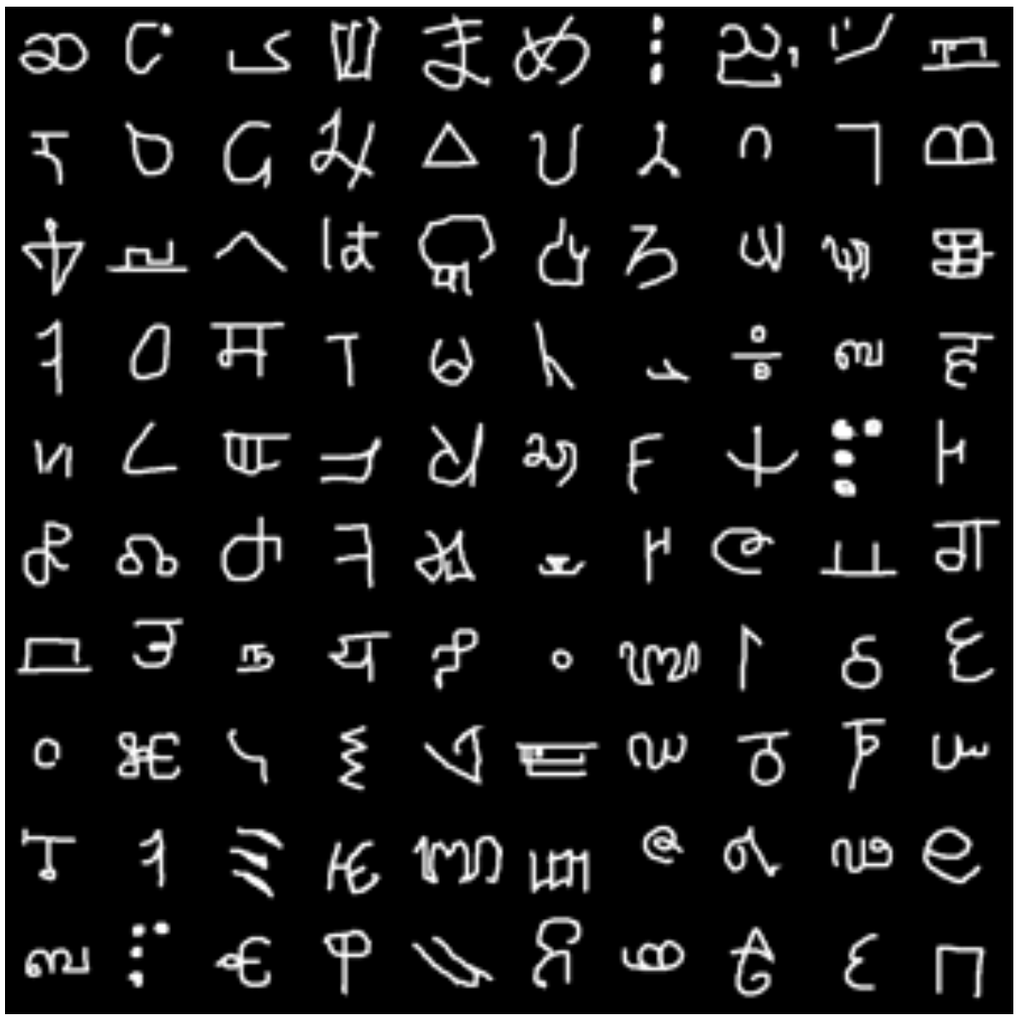


Salakhutdinov, 2015 Deep learning summer school talk

Challenge 2

Which one is written by human ?





Salakhutdinov, 2015 Deep learning summer school talk

Idea

Model the QMC data with a RBM, then sample from the RBM

Huang and LW, 1610.02746
PRB in press

Why is it useful ?

- When the fitting is perfect, we can completely avoid the “quantum” part of the QMC

- Even with an imperfect fitting, the RBM can still guide the QMC sampling



RBM is a recommender system
for the QMC simulations

Bonus: it is also fun to see
what it discovers for the physical model

Falicov-Kimball model

$$\hat{H}_{\mathrm{FK}} = \sum_{i,j} \hat{c}_i^{\dagger} \mathcal{K}_{ij} \hat{c}_j + U \sum_{i=1}^{N} \left(\hat{n}_i - \frac{1}{2} \right) \left(x_i - \frac{1}{2} \right)$$

SIMPLE MODEL FOR SEMICONDUCTOR-METAL TRANSITIONS: SmB_6 AND TRANSITION-METAL OXIDES

L. M. Falicov*

Department of Physics, University of California, Berkeley, California 94720

and

J. C. Kimball†

Department of Physics, and The James Franck Institute, University of Chicago, Chicago, Illinois 60637

(Received 12 March 1969)

We propose a simple model for a semiconductor-metal transition, based on the existence of both localized (ionic) and band (Bloch) states. It differs from other theories in that we assume the one-electron states to be essentially unchanged by the transition. The electron-hole interaction is responsible for the anomalous temperature dependence of the number of conduction electrons. For interactions larger than a critical value, a first-order semiconductor-metal phase transition takes place.

Falicov and Kimball, PRL1969

Collecting the QMC data

Probability distribution of the heavy atoms

$$p_{\mathrm{FK}}(\mathbf{x}) \sim e^{\frac{\beta U}{2}\sum_{i=1}^{N} x_i} \times \det\left(1 + e^{-\beta \mathcal{H}}\right)$$

$$\mathcal{H}_{ij} = \mathcal{K}_{ij} + \delta_{ij} U\left(x_i - 1/2\right) \qquad \text{scales as } \mathcal{O}(N^4)$$

cf, for the RBM

$$p(\mathbf{x}) \sim e^{\sum_{i=1}^{N} a_i x_i} \times \prod_{j=1}^{M} \left(1 + e^{b_j + \sum_{i=1}^{N} x_i W_{ij}}\right)$$

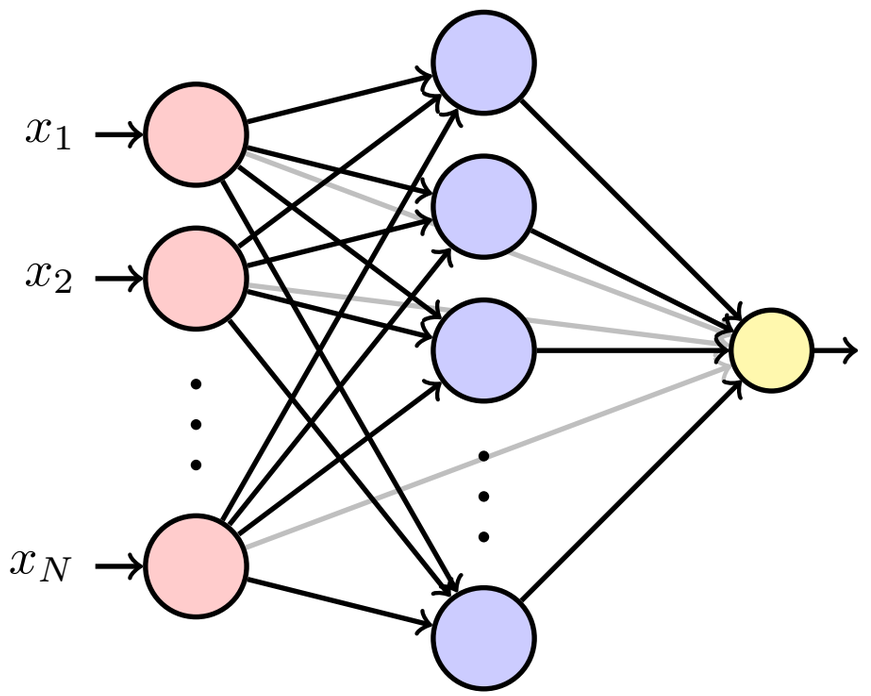
Train the RBM

$$p(\mathbf{x}) \sim p_{\mathrm{FK}}(\mathbf{x})$$

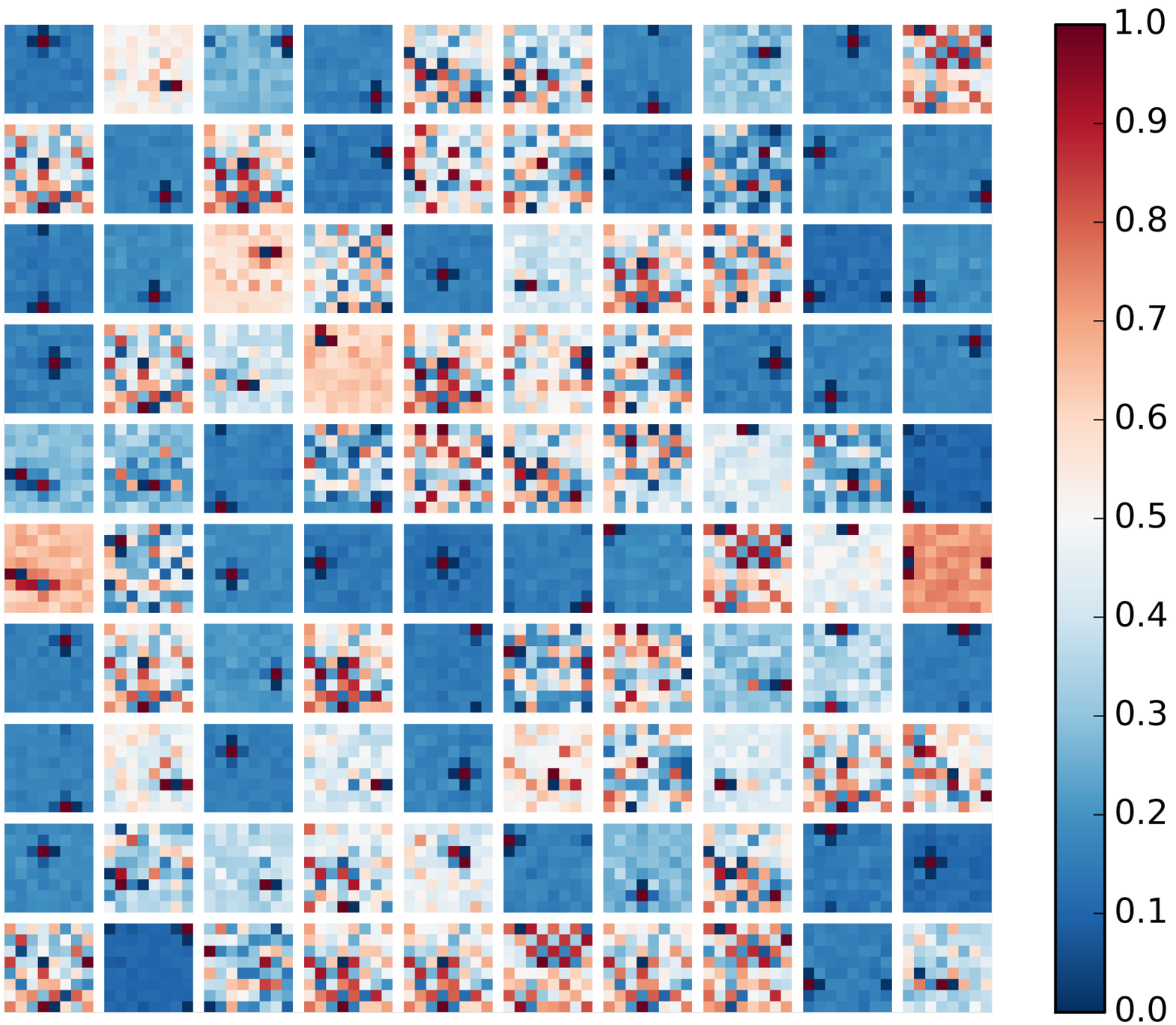
Supervised learning of the RBM

$$\sum_{i=1}^{N} a_i x_i + \sum_{j=1}^{M} \ln\left(1 + e^{b_j + \sum_{i=1}^{N} x_i W_{ij}}\right)$$

$$= \frac{\beta U}{2} \sum_{i=1}^{N} x_i + \ln \det\left(1 + e^{-\beta \mathcal{H}}\right)$$

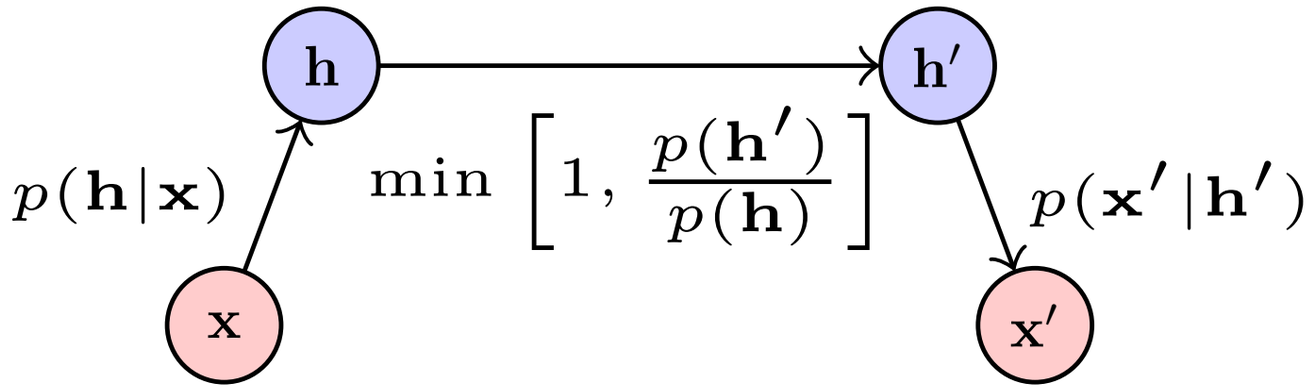


Trained Weight



Recommender updates

First simulate the RBM, then propose the move to QMC



$$A(\mathbf{x} \to \mathbf{x}') = \min\left[1, \frac{T(\mathbf{x}' \to \mathbf{x})}{T(\mathbf{x} \to \mathbf{x}')} \frac{p_{\mathrm{FK}}(\mathbf{x}')}{p_{\mathrm{FK}}(\mathbf{x})}\right]$$

$$A(\mathbf{x} \to \mathbf{x}') = \min\left[1, \frac{p(\mathbf{x})}{p(\mathbf{x}')} \frac{p_{\mathrm{FK}}(\mathbf{x}')}{p_{\mathrm{FK}}(\mathbf{x})}\right]$$

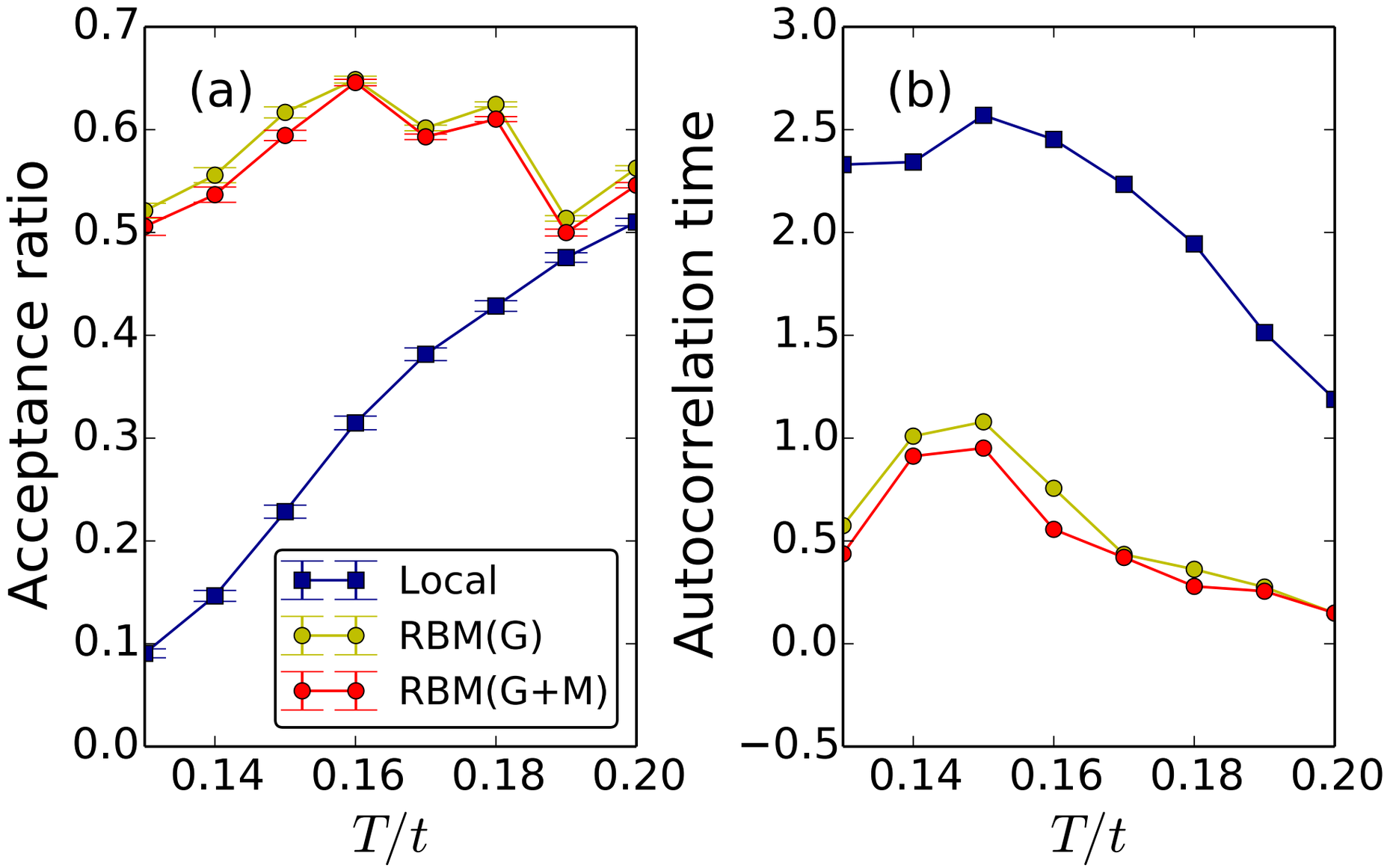
detailed balance condition for the RBM sampling

$$\frac{T(\mathbf{x} \to \mathbf{x}')}{T(\mathbf{x}' \to \mathbf{x})} = \frac{p(\mathbf{x}')}{p(\mathbf{x})}$$

The art of Monte Carlo methods

Huang and LW, 1610.02746, PRB in press
cf. Liu, Qi, Meng, Fu,1610.03137

It helps!



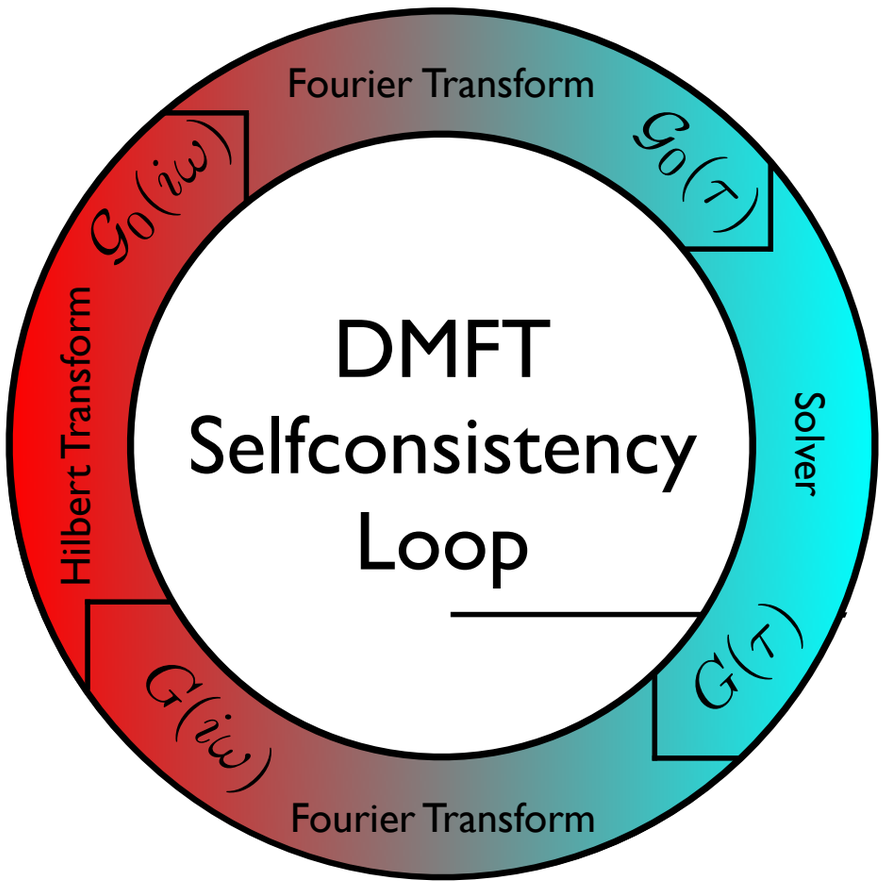
Falicov-Kimball model on 8*8 lattice

Another application

Recommender Engine for Quantum Impurity Solvers

Huang, Yang and LW, 1612.01871

Quantum impurity problems



Magnetic atoms hosted in metals

Quantum dots coupled to leads

Computational engine of DMFT

Kondo effect, heavy fermions physics…
doorway to strongly correlated electrons

$$S = -\int_0^\beta d\tau \int_0^\beta d\tau' \sum_{\sigma=\{\uparrow,\downarrow\}} c_\sigma^\dagger(\tau)\mathcal{G}_0^{-1}(\tau-\tau')c_\sigma(\tau')$$

$$+ U\int_0^\beta d\tau \left(n_\uparrow - \frac{1}{2}\right)\left(n_\downarrow - \frac{1}{2}\right)$$

Georges eta al, RMP 1996

Kotliar et al, RMP 2006

Continuous-time QMC solver

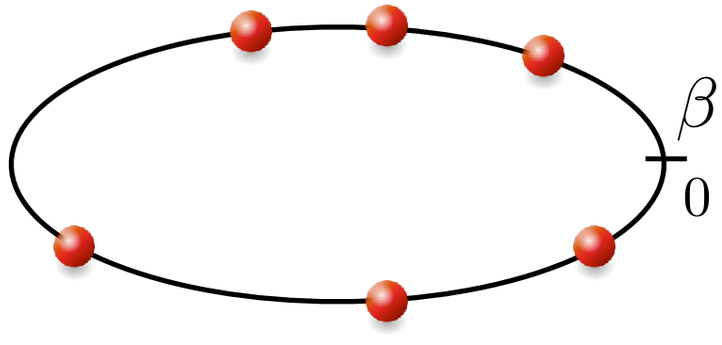
Gull et al, RMP 2011

$$Z/Z_0 = \sum_{k=0}^{\infty} \int_0^\beta d\tau_1 \int_{\tau_1}^\beta d\tau_2 \ldots \int_{\tau_{k-1}}^\beta d\tau_k \, (-U)^k |\det(G)|^2$$

$$= \sum_{\mathcal{C}} p(\mathcal{C})$$

$$\mathcal{C} = \{\tau_1, \tau_2, \ldots, \tau_k\}$$

$$G_{ij} = \mathcal{G}_0(\tau_i - \tau_j) - \delta_{ij}/2$$

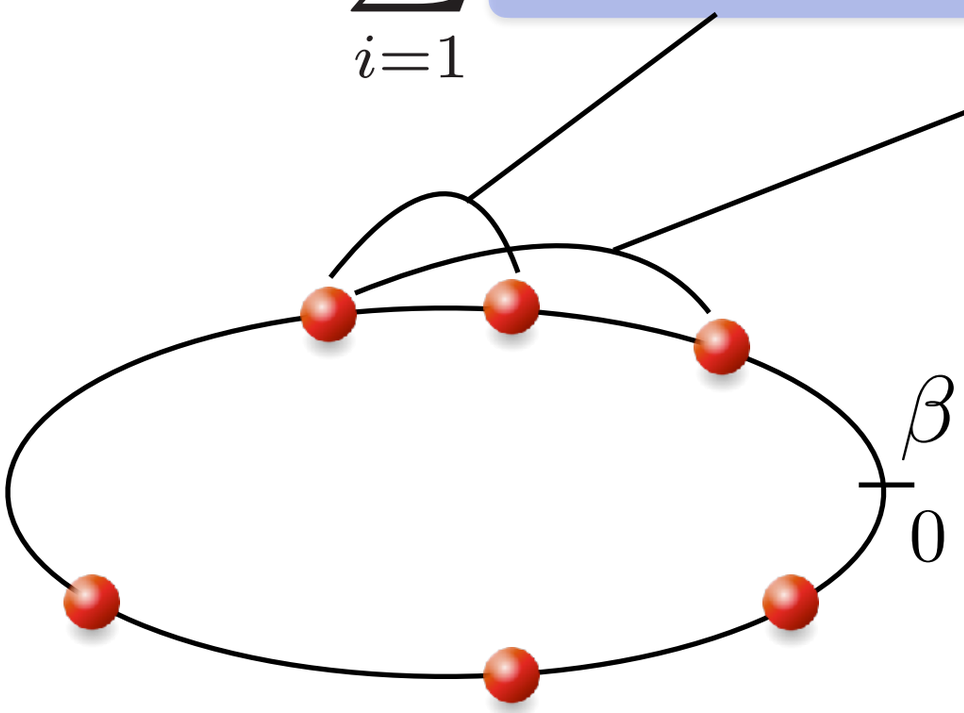


QMC as a **“classical molecular gas”** problem

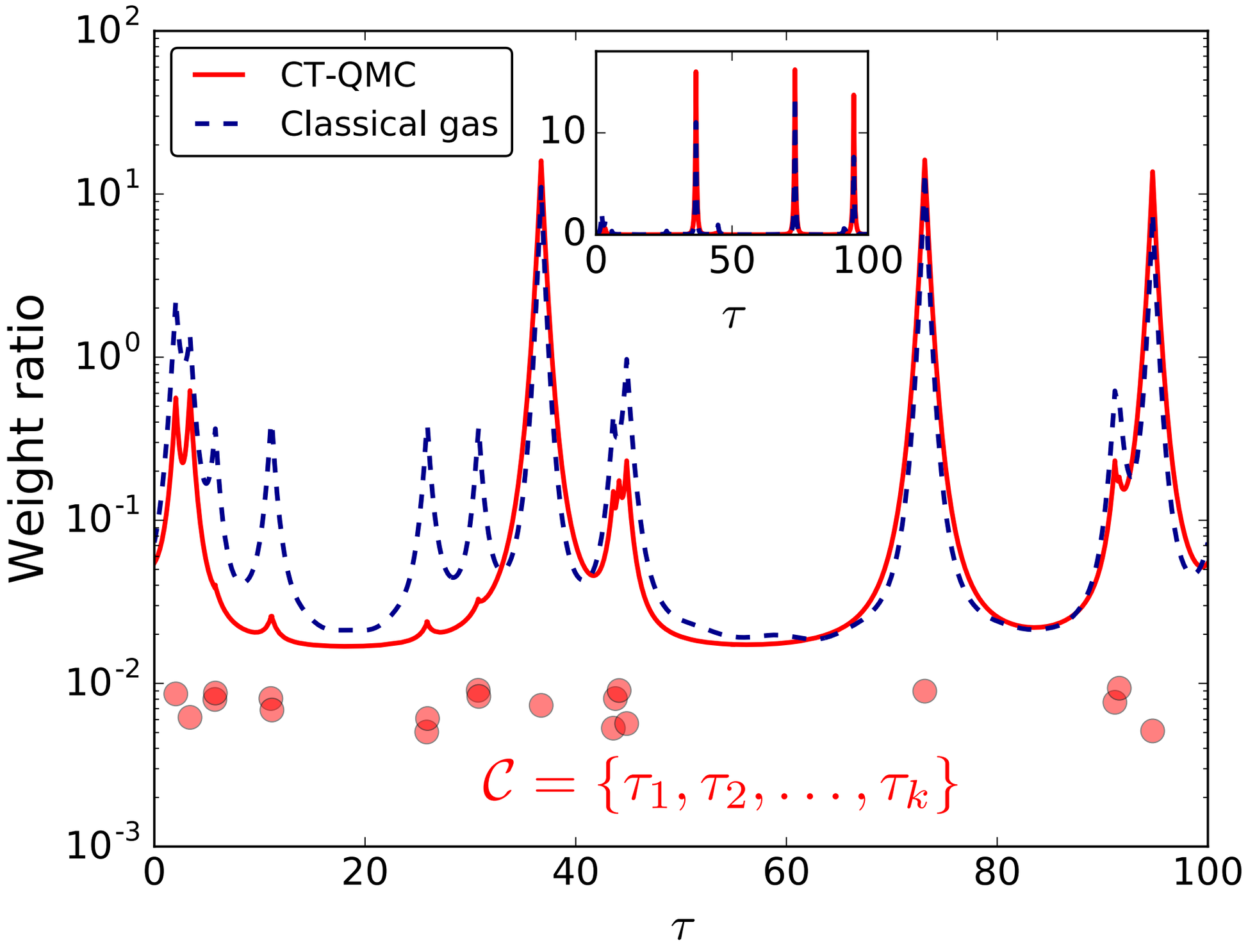
Reverse engineering the molecular gas energy

$$p(\mathcal{C}) \sim e^{-E(\mathcal{C})}$$

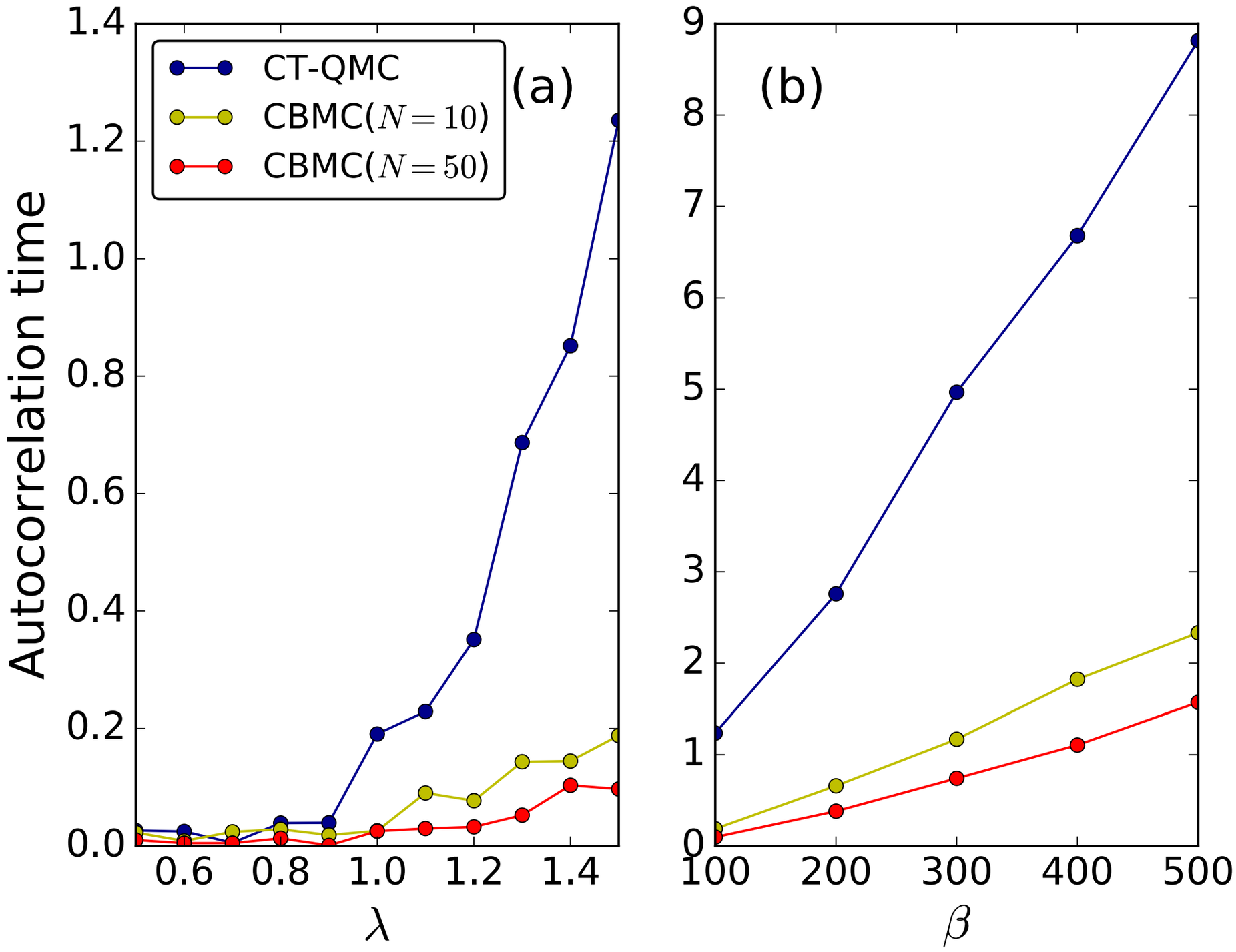
$$E(\mathcal{C}) = -\sum_{i=1}^{k} \mathcal{V}^{(2)}(\tau_{i+1} - \tau_i) - \sum_{i=1}^{k} \mathcal{V}^{(3)}(\tau_{i+1} - \tau_{i-1}) - \mu k$$



Effective 2-body attractive and 3-body repulsive interactions



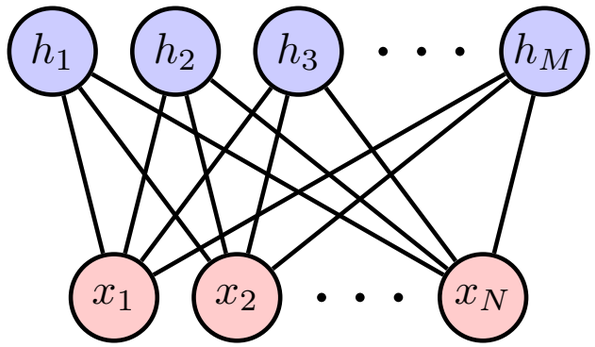
$$\frac{p(\mathcal{C} \cup \{\tau\})}{p(\mathcal{C})} \quad \text{V.S.} \quad \frac{e^{-E(\mathcal{C} \cup \{\tau\})}}{e^{-E(\mathcal{C})}}$$

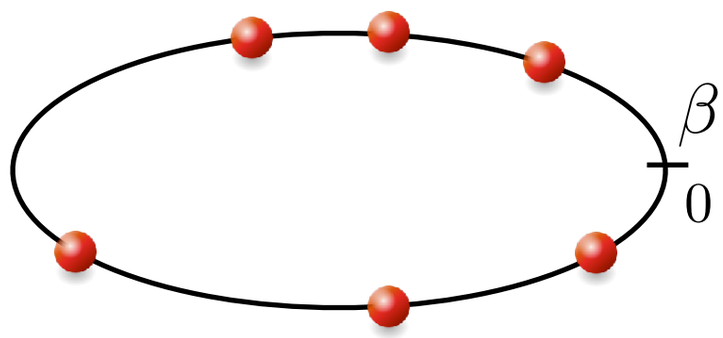


Molecular simulation techniques accelerate the quantum impurity solver

Recommender engines for QMC

Model: it is up to your imagination





Training: supervised learning of the statistical weight

Recommendation: detailed balance condition for unbiased physical results

Machine learning for many-body physics

- New tools for (quantum) many-body problems
- Moreover, it offers a new way of thinking
- Can we make new scientific discovery with it ?
- Can one design better algorithms with it ?



The fun just starts!

LW, 1606.00318

Li Huang and LW, 1610.02746

Li Huang, Yi-feng Yang and LW, 1612.01871